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ME 1060: DYNAMICS

Lecture 2: Planar kinematics of a particle

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Overview

- **2D motion of a particle - Rectangular cartesian coordinates**
- **2D motion of a particle – N-T coordinates**
- **2D motion of a particle - Polar coordinates**
- **Relative motion**

- **Supporting Reading Material**
  - Hibbeler, R.C. 2016, Engineering Mechanics Dynamics, 14th Edition, Pearson. (ISBN: 978-0-133-91538-9)
  - Meriam, J.L., and Kraige, L.G. 2008, Engineering Mechanics, Vol. 2: Dynamics, SI Unit, 6th Edition, John Wiley & Sons. (ISBN: 978-0-471-78703-7)

2D Motion of a particle - Rectangular Cartesian Coordinates

Occasionally the motion of a particle can best be described along a path that can be expressed in terms of its x, y, z coordinates. (z comes in when it is 3D motion)

Position

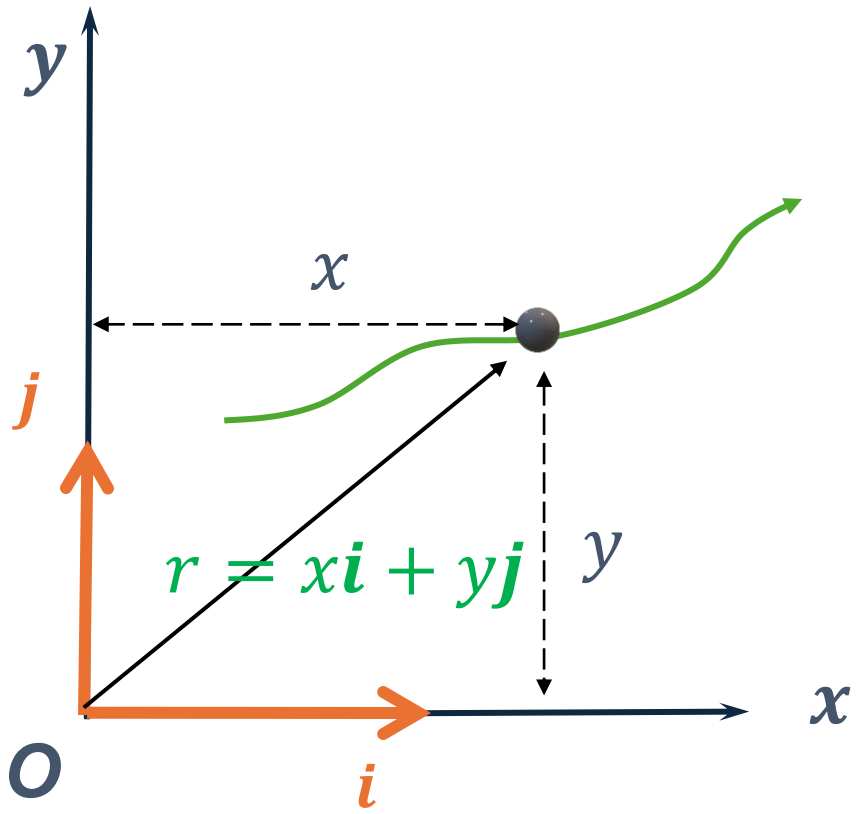
Position of the particle $\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j}$, is a vector

When the particle moves, the x, y components of r will be functions of time; i.e., $x = x(t), y = y(t)$, so that $\boldsymbol{r} = \boldsymbol{r}(t)$.

At any instant the magnitude of r is defined as

$$r = \sqrt{x^2 + y^2}$$

And the direction of $\boldsymbol{r}$ is specified by the unit vector $\boldsymbol{u_r} = \boldsymbol{r}/\, r$.



Velocity

The first time derivative of r yields the velocity of the particle. Hence,

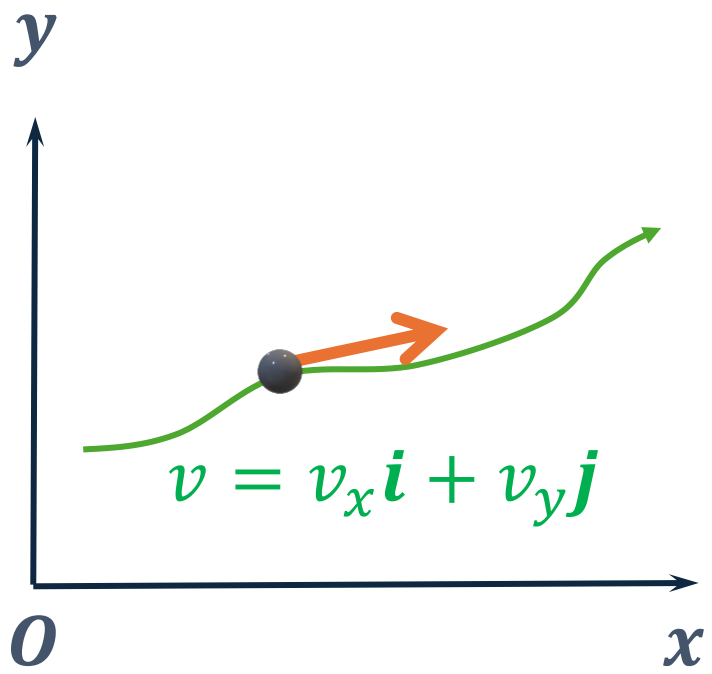
$$v = \frac{d\boldsymbol{r}}{dt} = \frac{d}{dt}(x\boldsymbol{i}) + \frac{d}{dt}(y\boldsymbol{j})$$

$$\boldsymbol{v} = \frac{d\boldsymbol{r}}{dt} = v_x\boldsymbol{i} + v_y\boldsymbol{j}$$

Where $v_x = \dot{x}$ and $v_y = \dot{y}$

Magnitude of the velocity $v = \sqrt{v_x^2 + v_y^2}$

Direction of the unit vector $\boldsymbol{u}_v = \boldsymbol{v}/v$



Acceleration

The first time derivative of r yields the velocity of the particle. Hence,

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = a_x\boldsymbol{i} + a_y\boldsymbol{j}$$

$a_x = \ddot{x}$ and $a_y = \ddot{y}$

$$a_y = \frac{dv_y}{dt} = \ddot{y} = v_y\frac{dv_y}{dy}$$

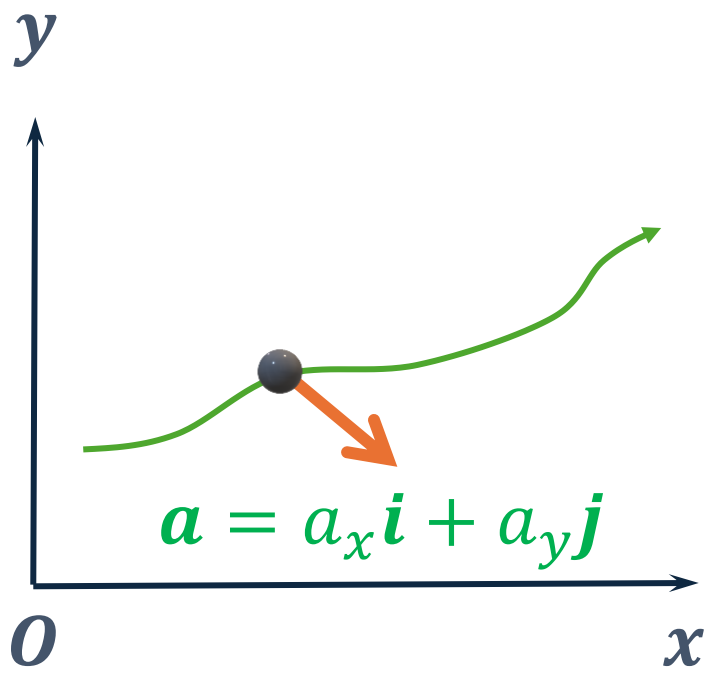
Magnitude of $\boldsymbol{a}$,

$$a = \sqrt{a_x^2 + a_y^2}$$

$$a_x = \frac{dv_x}{dt} = \ddot{x} = v_x\frac{dv_x}{dx}$$

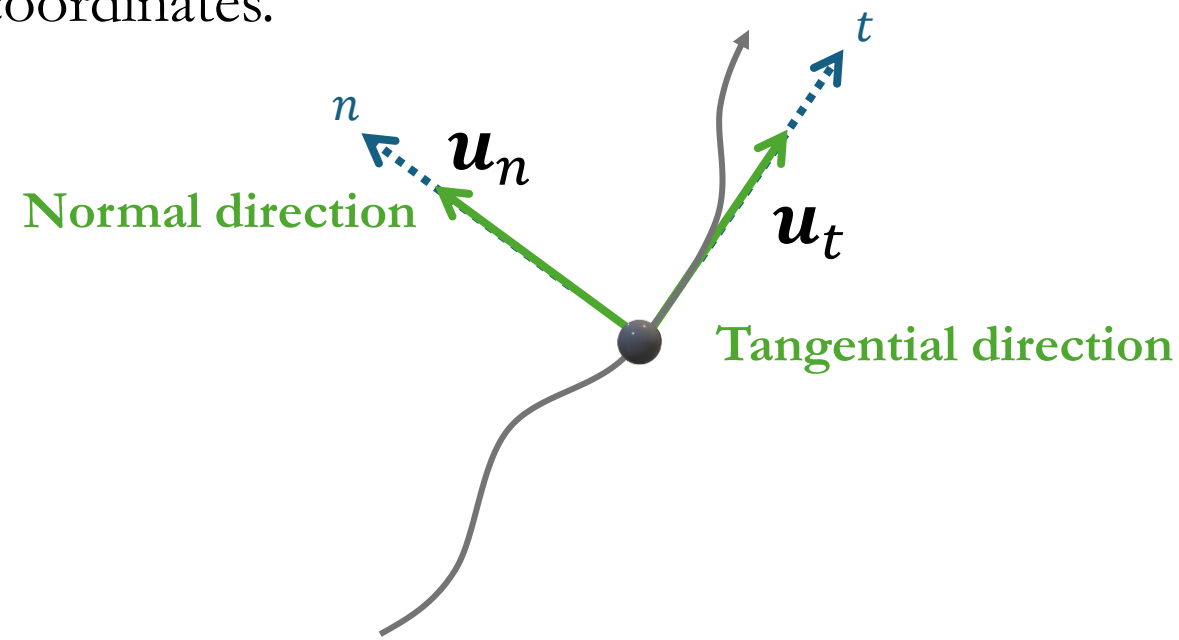
Direction of a,

Unit vector $u_a = \boldsymbol{a}/a$



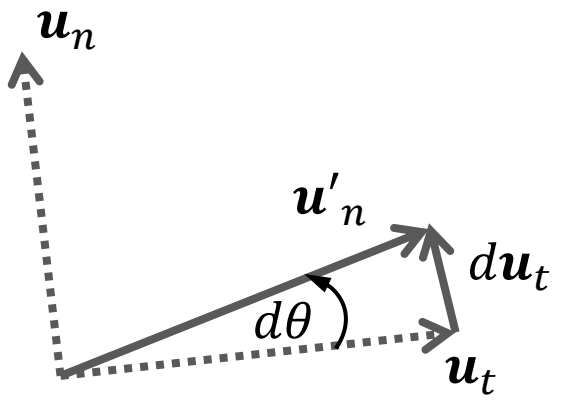
2D Motion of a particle - Normal and Tangential Components

When the path along which a particle travels is known, then it is often convenient to describe the motion using n and t coordinates.



The particle's velocity v has a direction that is always tangent to the path. Therefore,

$\boldsymbol{v} = v\boldsymbol{u}_t$ where $v = \dot{s}$



2D Motion of a particle - Normal and Tangential Components

Acceleration = $a = a_t u_t + a_n u_n$ where $a_t = \dot{v}$

and $a_n = \frac{v^2}{\rho}$ (ρ is the radius of curvature). How this comes?

The magnitude of u_t is always 1. Only its direction changes.For a small angular change dθ,

$$|d\mathbf{u}_t| \approx d\theta$$

and this change is directed toward the centre of curvature, i.e. in the normal direction u_n. Therefore,

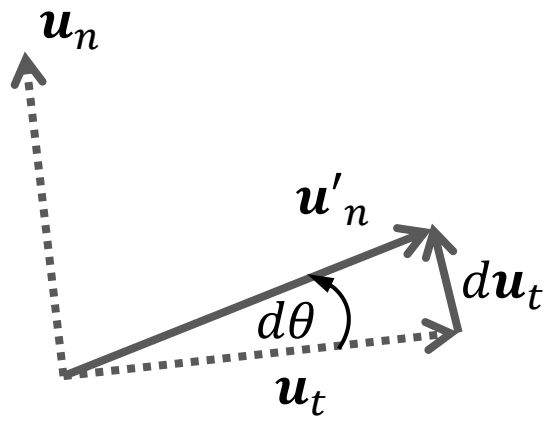
$$\frac{d\mathbf{u}_t}{dt} = \frac{d\theta}{dt}\mathbf{u}_n$$

Since $\frac{d\theta}{dt} = \frac{v}{\rho}$,

$$\frac{d\mathbf{u}_t}{dt} = \frac{v}{\rho}\mathbf{u}_n$$

Previously we got

$\boldsymbol{v} = v\boldsymbol{u}_t$



$$\mathbf{a} = \frac{dv}{dt}\mathbf{u}_t + v\frac{d\mathbf{u}_t}{dt}$$

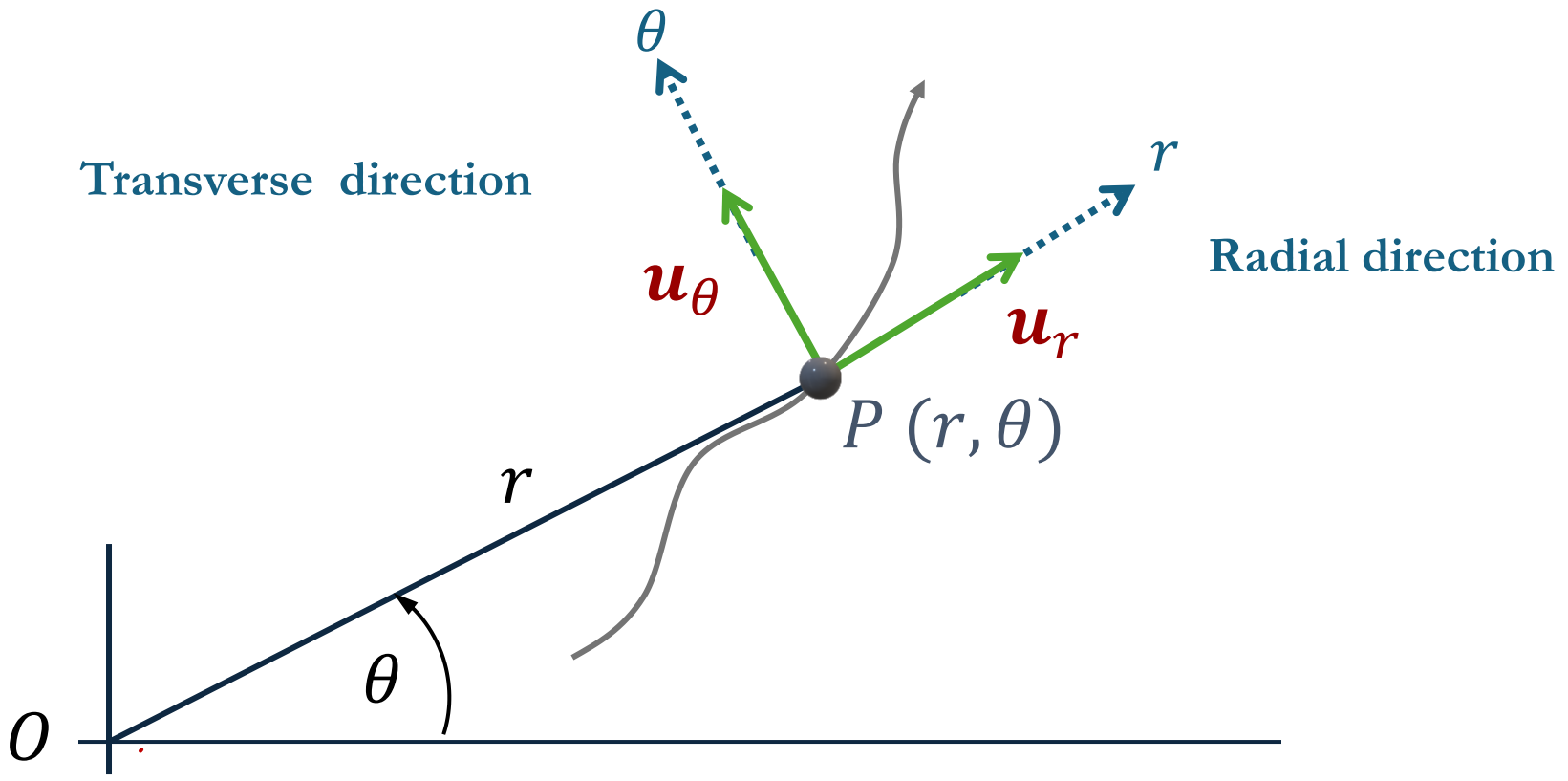
$$\frac{d\mathbf{u}_t}{dt} = \frac{v}{\rho}\mathbf{u}_n$$

$$\mathbf{a} = \frac{dv}{dt}\mathbf{u}_t + v\left(\frac{v}{\rho}\mathbf{u}_n\right)$$

$$\boxed{\mathbf{a} = \frac{dv}{dt}\mathbf{u}_t + \frac{v^2}{\rho}\mathbf{u}_n}$$

2D Motion of a particle - Polar Coordinates (r, θ)

We can specify the location of the particle using a radial coordinate r, which extends outward from the fixed origin O to the particle, and a transverse coordinate θ, which is the counterclockwise angle between a fixed reference line and the r axis.

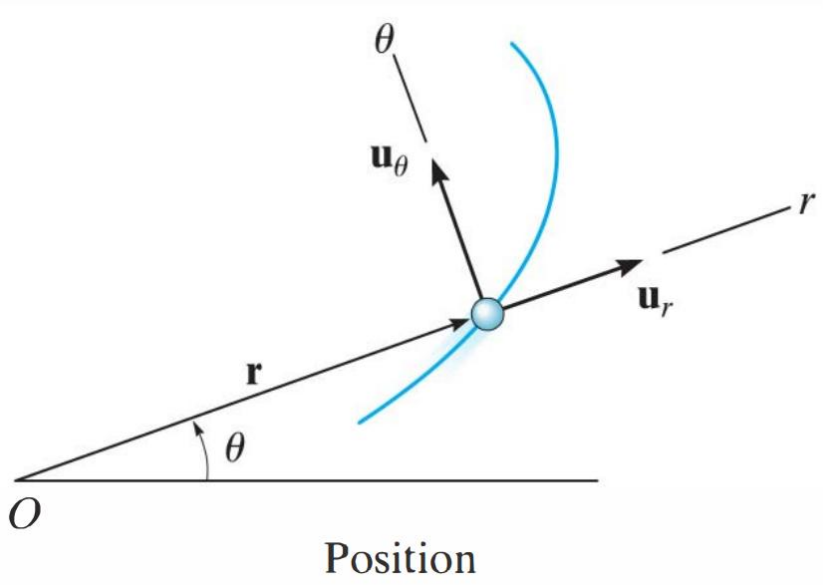


2D Motion of a particle - Polar Coordinates (r, θ)

Position

Position vector is defined by

$$\boldsymbol{r} = r\boldsymbol{u}_r$$



Position

Instantaneous Velocity

Instantaneous velocity $\boldsymbol{v}$

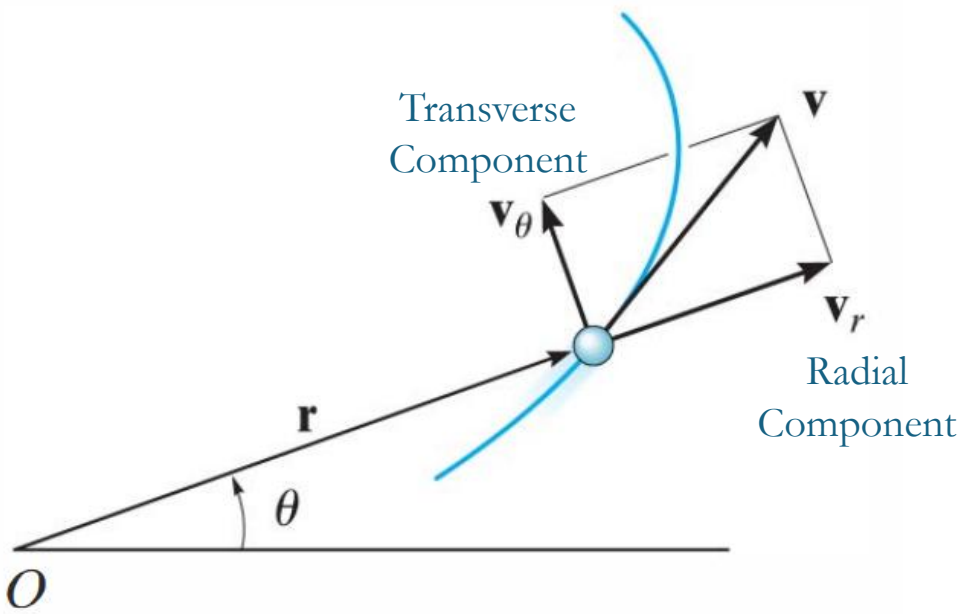
$$\boldsymbol{v} = \dot{\boldsymbol{r}} = \dot{r}\boldsymbol{u_r} + r\dot{\boldsymbol{u}}_{\boldsymbol{r}}$$

Where $\dot{\boldsymbol{u}}_r = \dot{\theta}\boldsymbol{u}_\theta$

Substituting to the above equation:

$$\boldsymbol{v} = v_r\boldsymbol{u_r} + v_\theta\boldsymbol{u_\theta}$$

Where $v_r = \dot{r}$ and $v_\theta = r\dot{\theta}$



The magnitude of $v = \sqrt{(\dot{r})^2 + \left(r\dot{\theta}\right)^2}$ and the direction of $\boldsymbol{v}$ is tangential to the path.

Velocity Components

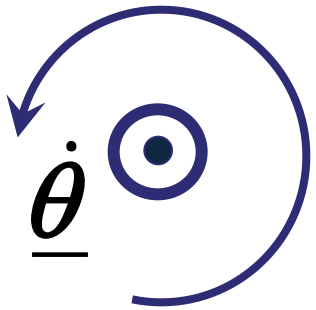
$$v_\theta = r\,\dot{\theta}$$

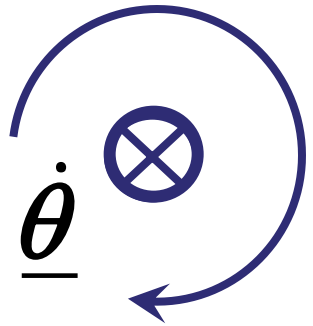
Transverse component

$$v_r = \dot{r}$$

Radial component

Right hand rule (notation) for rotary vectors

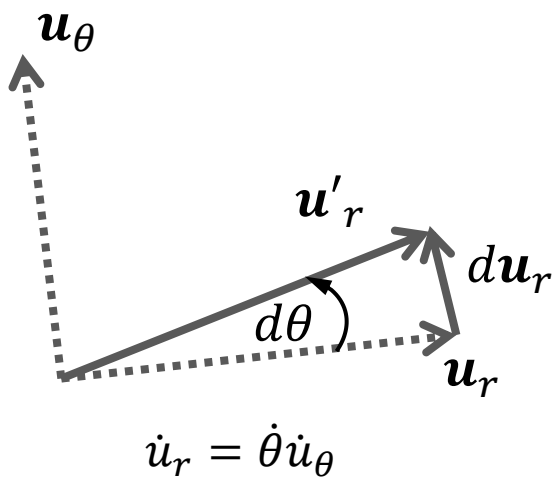




Acceleration

$$\boldsymbol{v} = \dot{r}\boldsymbol{u_r} + r\dot{\theta}\boldsymbol{u_\theta}$$

Acceleration $\boldsymbol{a} = \dot{\boldsymbol{v}}$



$$\dot{u}_r = \dot{\theta}\dot{u}_\theta$$

$$\boxed{d\mathbf{u}_r = d\theta\,\mathbf{u}_\theta}$$

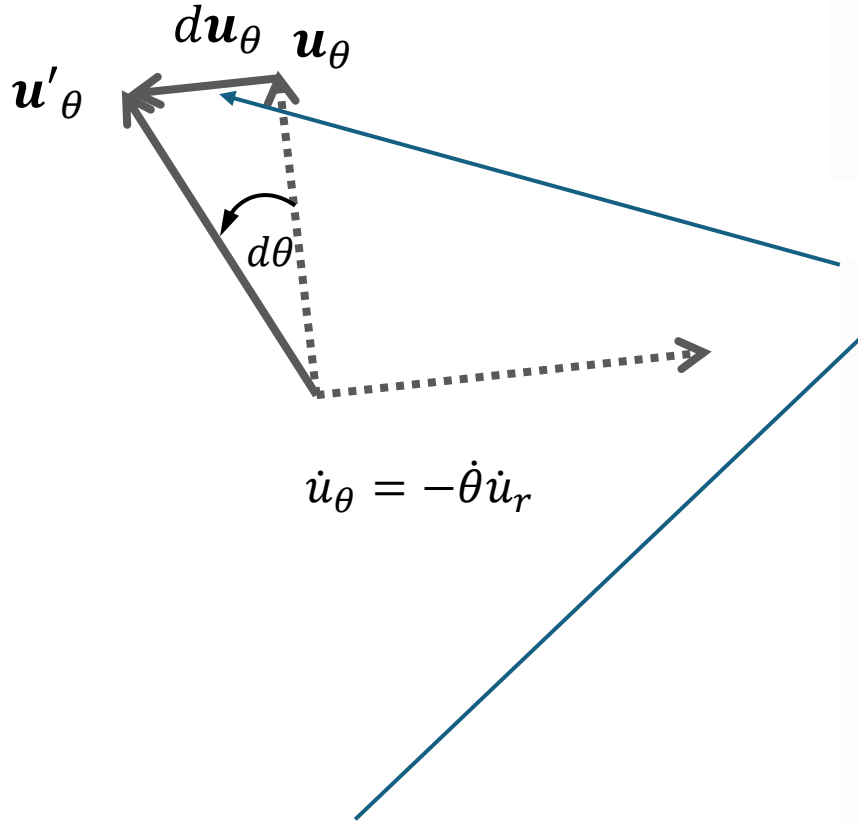
$$\frac{d\mathbf{u}_r}{dt} = \frac{d\theta}{dt}\mathbf{u}_\theta$$

$$\dot{\theta} = \frac{d\theta}{dt},$$

$$\boxed{\dot{\mathbf{u}}_r = \dot{\theta}\mathbf{u}_\theta}$$

$$d\mathbf{u}_\theta = \mathbf{u}'_\theta - \mathbf{u}_\theta$$

$$|d\mathbf{u}_\theta| = d\theta$$



$$d\mathbf{u}_\theta = -d\theta\,\mathbf{u}_r$$

$$\dot{u}_\theta = -\dot{\theta}\dot{u}_r$$

$$\frac{d\mathbf{u}_\theta}{dt} = -\frac{d\theta}{dt}\mathbf{u}_r$$

$$\boxed{\dot{\mathbf{u}}_\theta = -\dot{\theta}\mathbf{u}_r}$$

Negative direction

u_r

Acceleration

Acceleration $\boldsymbol{a}$

$$\boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{r}\boldsymbol{u_r} + \dot{r}\dot{\boldsymbol{u}}_{\boldsymbol{r}} + \dot{r}\dot{\theta}\boldsymbol{u}_{\theta} + r\ddot{\theta}\boldsymbol{u}_{\theta} + r\,\dot{\theta}\dot{\boldsymbol{u}}_{\theta}$$

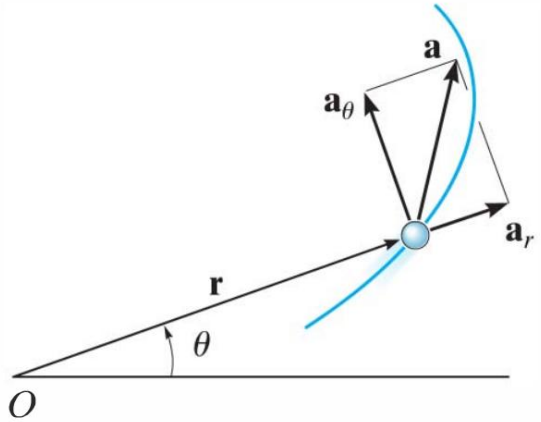
$$\dot{\boldsymbol{u}}_{\boldsymbol{\theta}} = -\dot{\theta}\mathbf{u}_{\mathbf{r}}$$

Therefore,

$$a = a_r u_r + a_\theta u_\theta$$

Where

$$a_r = \ddot{r} - \mathrm{r}\,\dot{\theta}^2$$
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

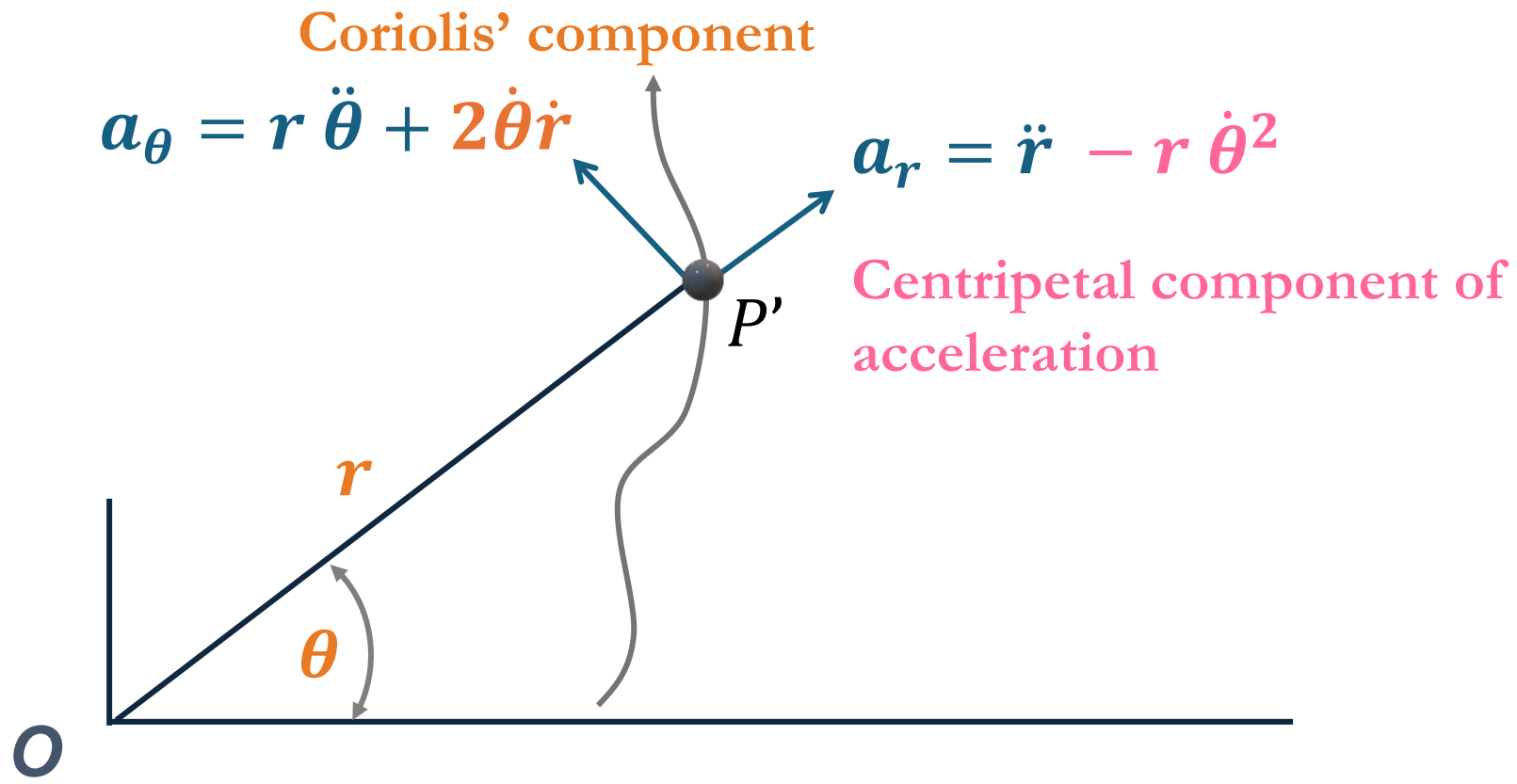


The magnitude of $\boldsymbol{a} = \sqrt{\left(\ddot{r} - \mathrm{r}\,\dot{\theta}^2\right)^2 + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)^2}$

The direction is determined from the vector addition of its two components. It will ***not*** be tangential to the path.

2D Motion of a particle - Planar-Polar Coordinates

Acceleration Components



2D Motion of a particle - Planar-Polar Coordinates

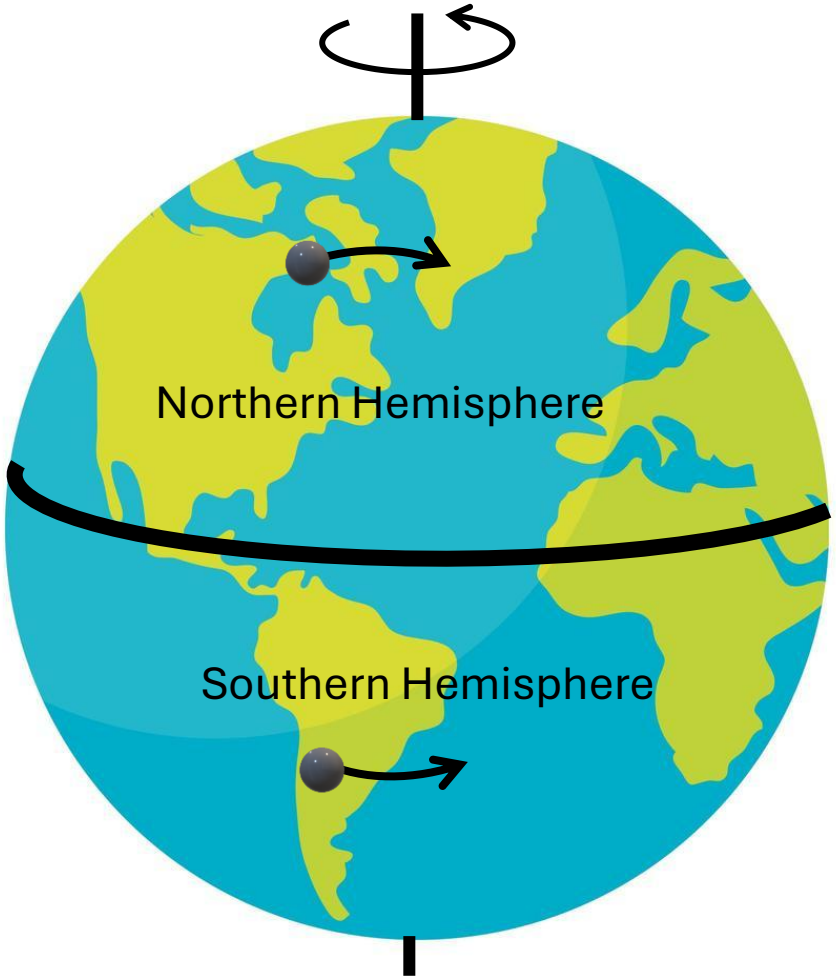
Acceleration Components

Coriolis' component of acceleration $= 2\dot{\theta}\dot{r}$

Centripetal comp. of acceleration $= -r\dot{\theta}^2$
$= \dot{\theta}(\dot{\theta})r$

Coriolli's compt. Effect

Due to the earth's rotation

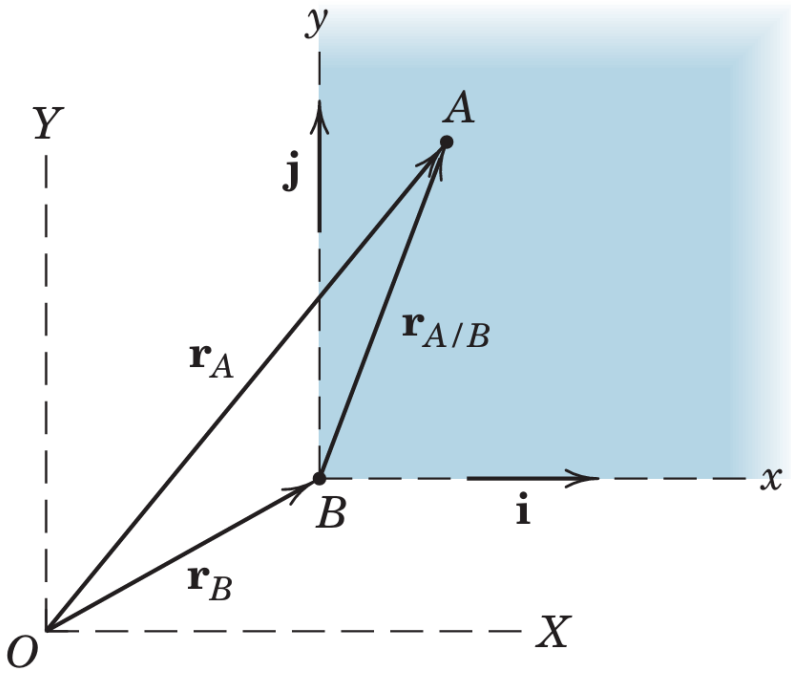


Objects deflect to the right in the northern hemisphere

Objects deflect to the left in the southern hemisphere

The Coriolis Effect Explained (youtube.com)

Relative Motion



The unit vectors along the x- and y −axes are i and j, and x and y are the coordinates of A measured in the $x - y$ frame.

$$r_A = r_B + r_{A/B}$$

Taking the time derivative:

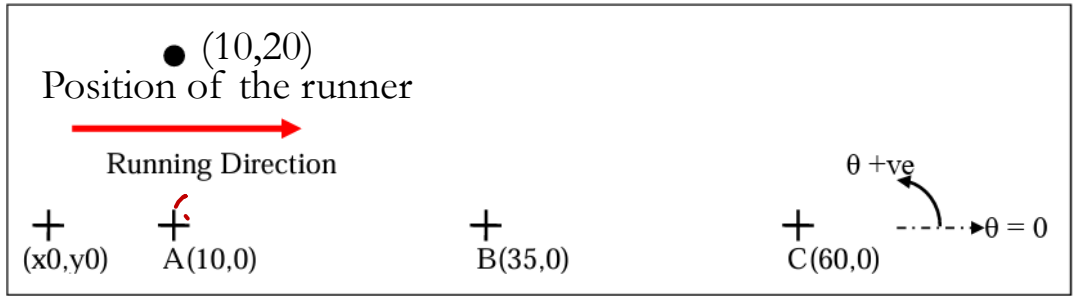
$$v_A = v_B + v_{A/B}$$

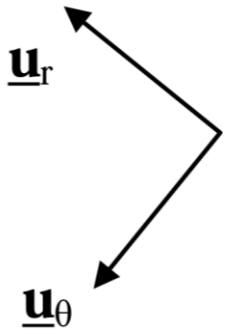
Taking the second time derivative

$$a_A = a_B + a_{A/B}$$

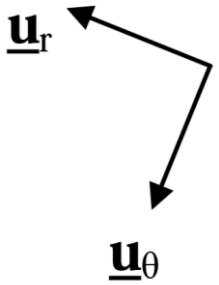
Exercise 1

The position of a runner during sprint interval training is shown below. Observers (coaches) are located at A, B and C, 20m to the side of the track. Match the corresponding $r - \theta$ directions for each observer.

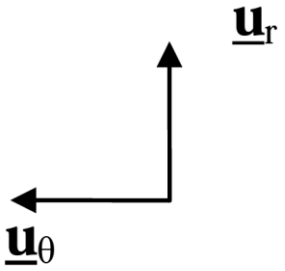




(a)



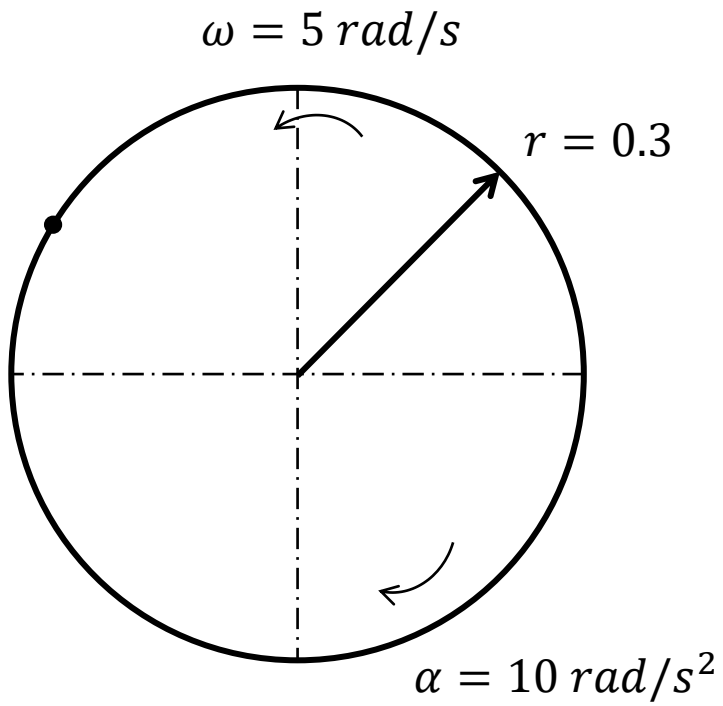
(b)



(c)

Example 1

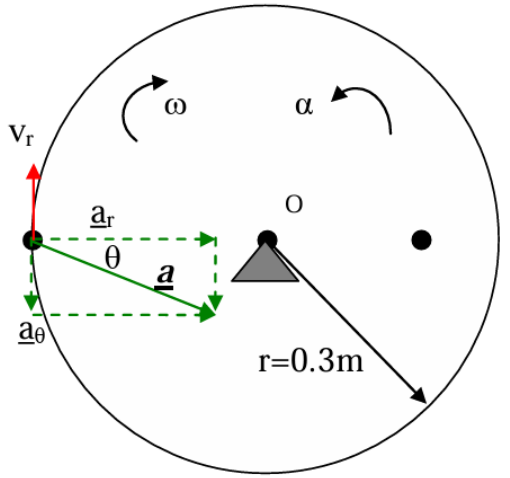
At a given instant, a fixed wheel of diameter $0.6m$, has an angular velocity of $5\ rad/s$ *anti*clockwise and an angular deceleration of $10\ rad/s^2$. Find the acceleration of a point on the rim.



Example 1

At a given instant, a fixed wheel of diameter $0.6m$, has an angular velocity of $5\ rad/s$ clockwise and an angular deceleration of $10\ rad/s^2$. Find the acceleration of a point on the rim.

Solution:



When $r = \ 0.3,\ \dot{r} = \ddot{r} = 0$

$\omega = 5\ rad/s$ and $\alpha = -10\ rad/s^2$

$a = \left(\ddot{r}\ - \mathrm{r}\ \dot{\theta}^2\right)u_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)u_\theta$

$a_r = -0.3 \times 5^2 = 7.5\ m/s^2$ towards the centre of the rotation

$a_\theta = 0.3 \times (-10) = 3\ ms^{-2}$ perpendicular to the radius, backwards relative to the tangential motion.

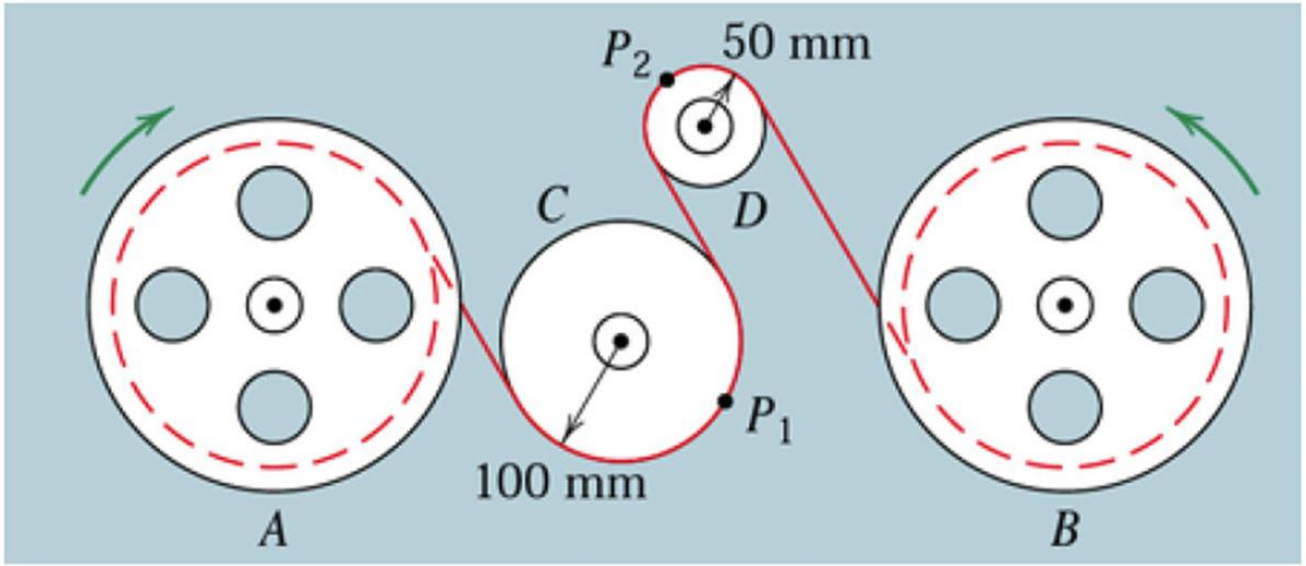
$$|a| = \sqrt{a_r^2 + a_\theta^2} = \sqrt{65.25} = 8.1\ ms^{-1}$$

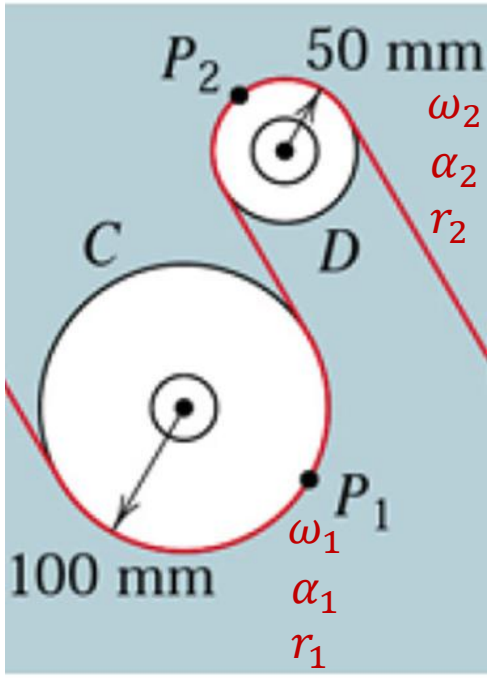
$$\theta = \mathrm{atan}\left(\frac{3}{7.5}\right) = 21.8^o$$

Example 2

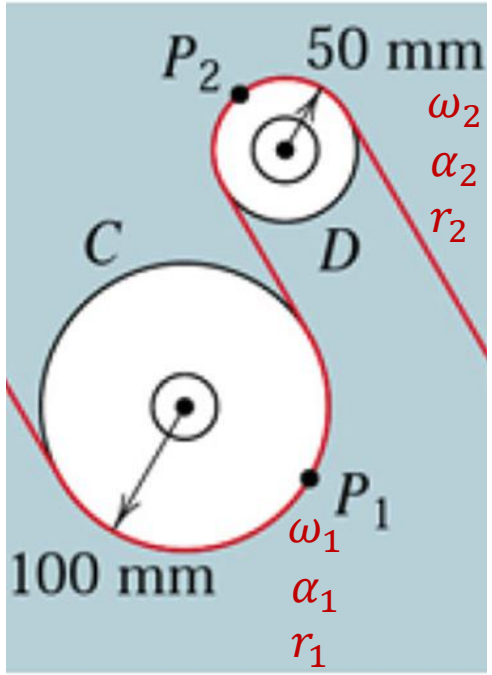
Figure shows a reel of magnetic tape being passed from reel A to reel B with idler pulleys C and D. Calculate the speed of the tape and the total acceleration of the tape at points P1 and P2. Measurements indicate that the normal component of acceleration at P1 is $40\ m/s^2$ and the tangential component of acceleration at P2 is $30\ m/s^2$.

$$a = -(\ddot{r} - r\,\dot{\theta}^2)u_n + (r\ddot{\theta} + 2\dot{r}\dot{\theta})u_t$$





Example 2

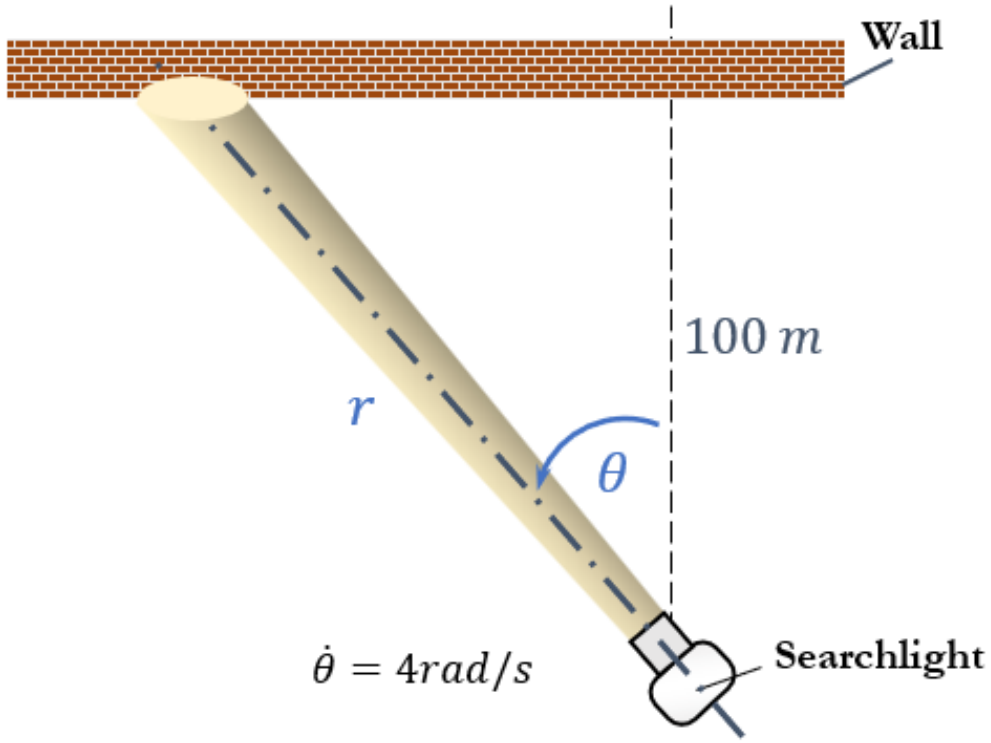


[Ans. v_{P1} = 2 m/s, P1 acceleration = $50\ m/s^2$, P2 acceleration = $85.4\ m/s^2$]

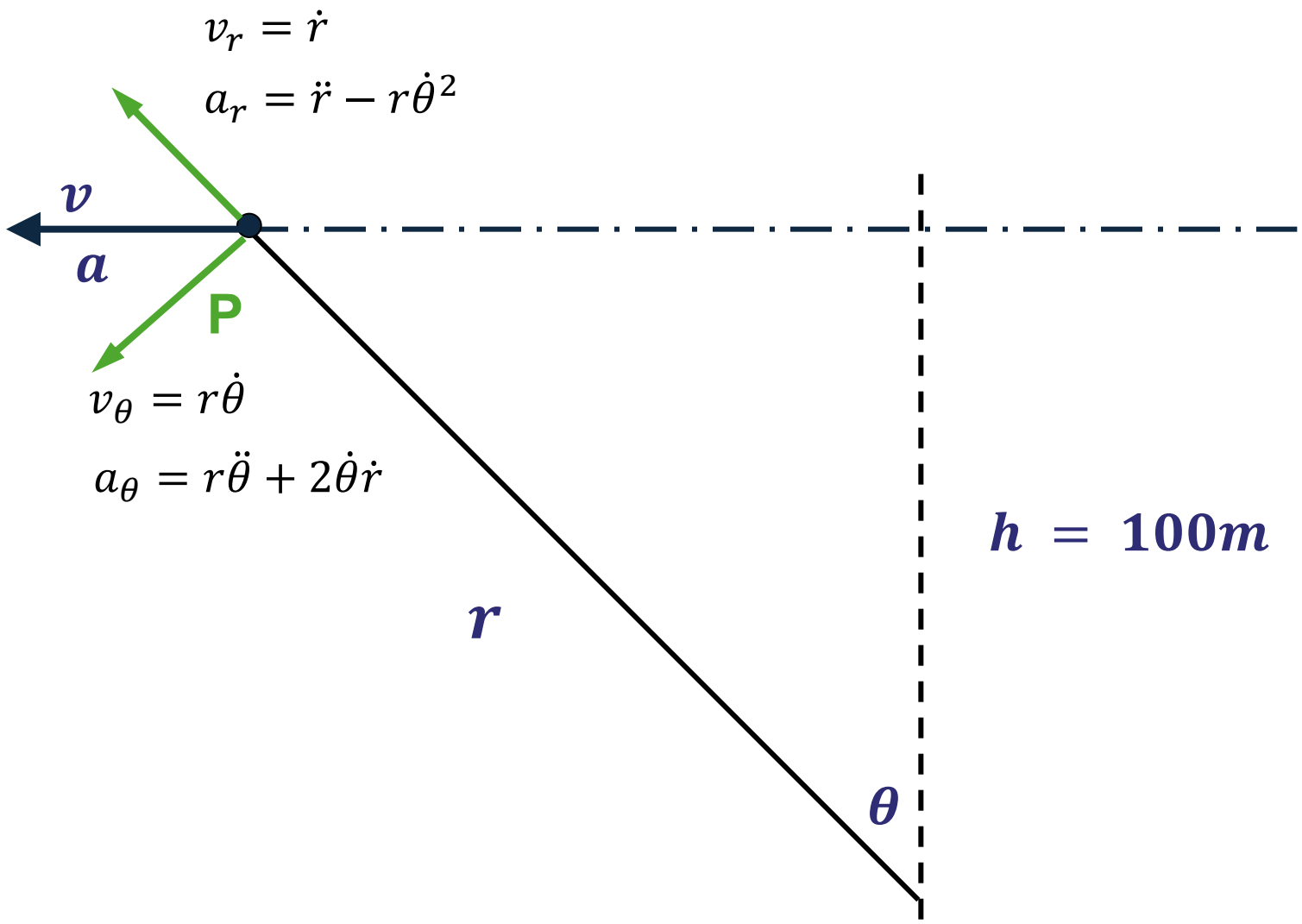
Example 3

The searchlight shown in Figure 1 casts a spot of light along the face of a wall that is located $100m$ from the searchlight. Determine the magnitude of the velocity and acceleration at which the spot appears to travel across the wall at the instant $\theta = 45^o$.

The searchlight is rotating at a constant rate of $4rad/s$.



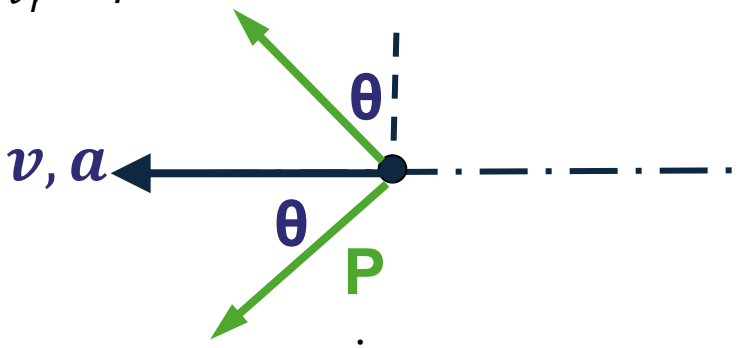
Example 3


$v_r = \dot{r}$
$a_r = \ddot{r} - r\dot{\theta}^2$
v
a
P
$v_\theta = r\dot{\theta}$
$a_\theta = r\ddot{\theta} + 2\dot{\theta}\dot{r}$
$h = 100m$
r
θ

Example 3

$a_r = \ddot{r} - r\dot{\theta}^2$

$v_r = \dot{r}$



$v_\theta = r\dot{\theta}$

$a_\theta = r\ddot{\theta} + 2\dot{\theta}\dot{r}$

↑ $\left(\ddot{r} - r\dot{\theta}^2\right)\cos\theta = \left(r\ddot{\theta} + 2\dot{\theta}\dot{r}\right)\sin\theta\,,$

$\ddot{\theta} = 0$ (no acceleration in)

$\ddot{r} = 2\dot{\theta}\dot{r}.\tan\theta + r\dot{\theta}^2$

← $a = \left(\ddot{r} - r\dot{\theta}^2\right)\sin\theta + \left(r\ddot{\theta} + 2\dot{\theta}\dot{r}\right)\cos\theta\,, \ddot{\theta} = 0$

$a = \underline{6400 m/s^2}$

Since there are no velocity and acceleration components normal to the wall;

↑ $\dot{r}\cos\theta = r\dot{\theta}\sin\theta$

$\dot{r} = r\dot{\theta}\tan\theta$

← $v = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$

$= \underline{800 m/s}$

Example 3

Solution (analytical)

$$r\cos\theta = h = 100m, r = h(\sec\theta),$$

$$\dot{r} = h\left(\sec\theta\,.\tan\theta\,.\dot{\theta}\right)$$

$$\ddot{r} = h.\begin{Bmatrix}\dot{\theta}\left[\sec\theta\,.\sec^2\theta\,.\dot{\theta} + \tan\theta\,.\sec\theta\,.\tan\theta\,.\dot{\theta}\right]\\ +(\sec\theta\,.\tan\theta)\ddot{\theta}\end{Bmatrix}$$

$$\dot{\theta} = 4rad/s = cont. \Rightarrow \ddot{\theta} = 0$$

$$at\theta = 45^{0,}, r = 100.\sec 4\,5^0 = 141.42m$$

$$\dot{r} = 565.68m/s$$

$$\ddot{r} = 6{,}788.2m/s^2$$

Example 3

$v_r = \dot{r} = 565.68 m/s, \rightleftarrows$

$v_\theta = r\dot{\theta} = 141.42 \times 4 m/s = 565.68 m/s$

$v = \sqrt{{v_r}^2 + {v_\theta}^2} = \underline{800 m/s}$

$a_r = \ddot{r} - r\dot{\theta}^2 = (6{,}788.2 - 141.42 \times 4^2) m/s^2$

$= 4{,}525.5 m/s^2$

$a_\theta = r\ddot{\theta} + 2\dot{\theta}\dot{r} = 0 + 2 \times 4 \times 565.68 = 4{,}525.44 m/s^2$

$a = \sqrt{{a_r}^2 + {a_\theta}^2} = \underline{6400 m/s^2}$

Home Work

A racing car Q is moving at a constant speed of $150\ km/hr$ along a circular track of radius $300\ m$ and centre O, as shown in Figure-3. The car is being filmed from a camera mounted on a tripod at the fixed point P as shown. At the instant depicted in the figure, $\theta = 30\text{deg}$. For this instant, determine the angular velocity and angular acceleration of the camera so that the image of Q remains centrally positioned in the viewfinder.

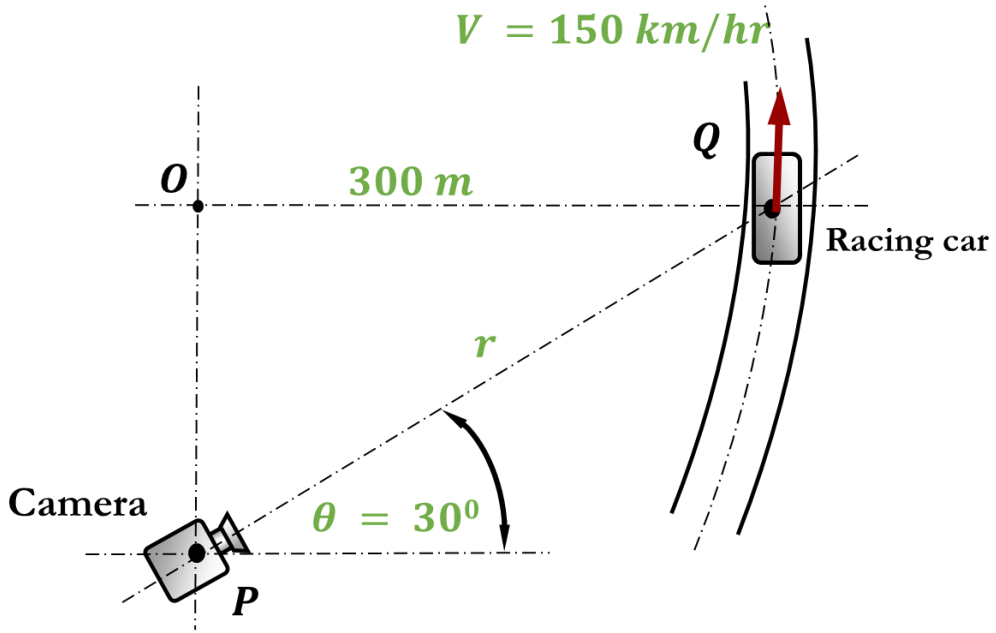


Figure-3

Home Work

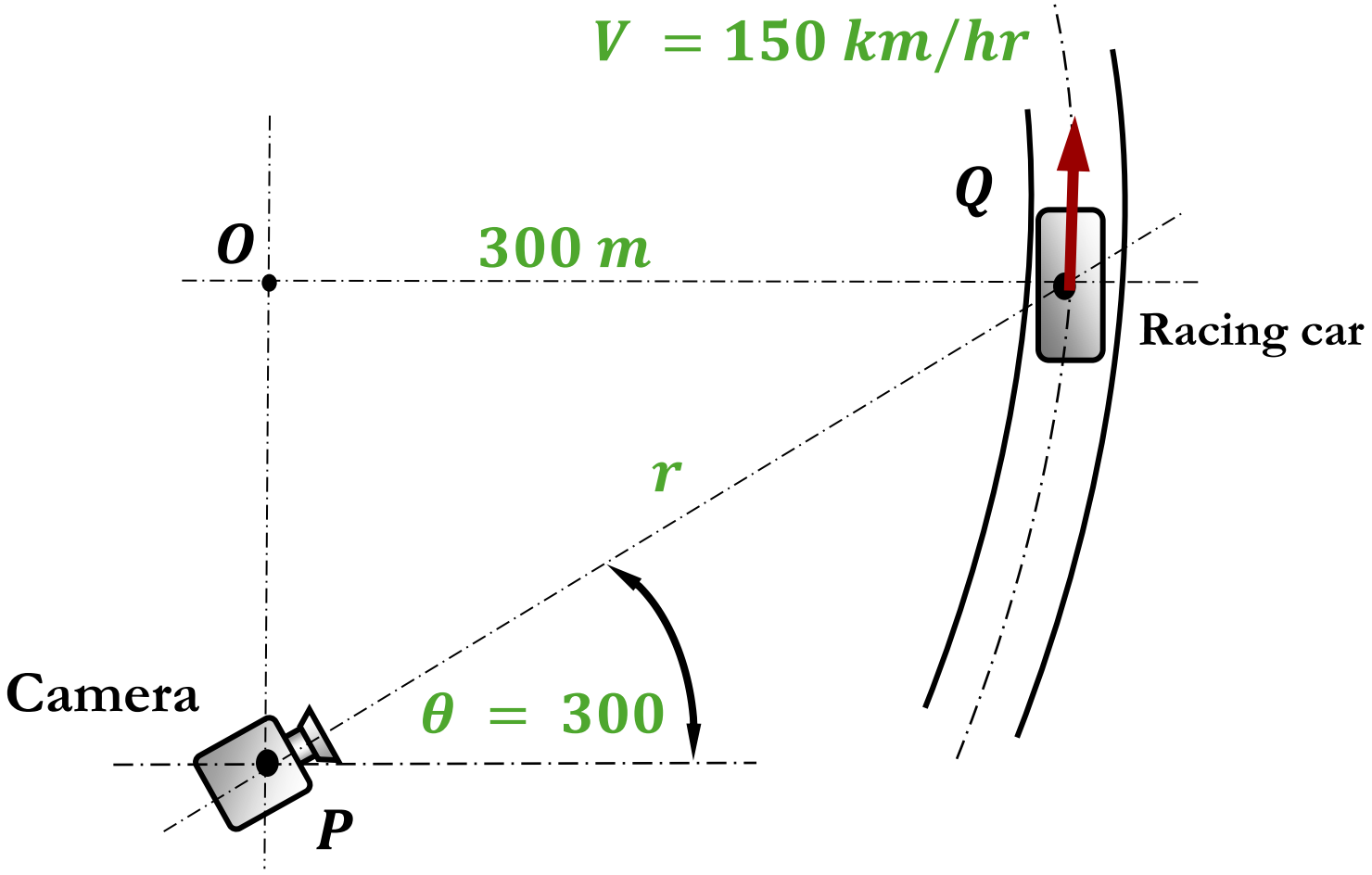


Figure-3